

text

- The text element adds text. The text can be selected using the mouse. x and y define the starting point of the text

attributes	Stroke Type & Description
1	stroke – defines color of text, line or outline of any element.
2	stroke-width – defines thickness of text, line or outline of any element.
3	stroke-linecap – defines different types of ending of a line or outline of any path.
4	stroke-dasharray – used to create dashed lines

circle

- Define a circle. cx and cy are the center coordinates, and r is the radius. fill is a common attribute and represents the figure color.

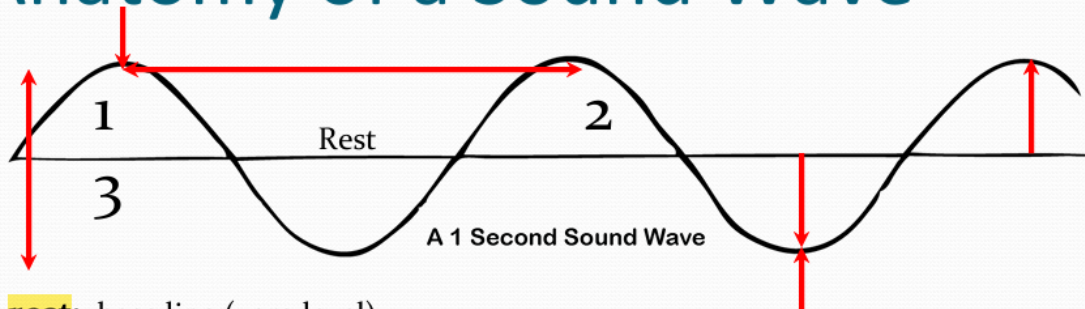
rect

- Defines a rectangle. x, y are the starting coordinates, with width and height.

line

- x_1 and y_1 define the starting coordinates. x_2 and y_2 define the ending coordinates. stroke is a common attribute and represents the line color.

Anatomy of a Sound Wave



rest: base line (zero level)

Crest: Point that exhibits **maximum amount of upward** displacement from the rest position.

Trough: Point that exhibits **maximum amount of downward** displacement from the rest position.

Wavelength: Distance from **crest** to **crest**.

Frequency: Number of waves to pass a point in one second. (How **Fast**)

Amplitude: Distance from **rest to crest** or **rest to trough**.

Pitch: Quality of a sound determined by the rate of vibration. The degree of highness or lowness of a tone. (How **High**)

- **Sample Rate:** The rate or frequency at which a sound is **sampled to digitize it**. Measurements of a sound wave over time measured in **Hz (cycles/sec)**.

- **Bit Rate** (Depth) (bits/seconds): The number of computer bits that are conveyed or processed per unit of time.
 - 48 kHz: standard for digital audio
 - CD audio is sampled at 44.1 kHz
- **Waveform**: a visual representation of an audio signal (sound clip).
- **Clip**: is a short segment of audio. It can be combined with others to make an audio track.
- **Track** (Channel): an area upon which audio clips move. It may be a single waveform or could be made up of clips.
 - Single Channel=**Mono**
 - Two Channels=**Stereo** (use both ears)
 - **Mono** = recorded with one microphone
 - **Stereo** = recorded with two microphones
 - **5.1** = recorded with six microphones (.1 is the bass (main) channel)

- **DX, FOL, SFX** should be **mono** files.
 - **MX and BGs** should be **stereo** files.
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- **Omnidirectional**
Picks up sound evenly from **all directions** (omni means "all" or "every")
 - **Unidirectional**
Picks up sound predominantly **from one direction**. This includes cardio and hypercardioid microphones
 - **Bidirectional**
Picks up sound from **two opposite directions**.

Common Audio Formats

- | | |
|---|---|
| <ul style="list-style-type: none">● WAV (.wav)<ul style="list-style-type: none">● Developed by IBM and Microsoft for Windows● High Quality (Lossless)● Large File Size● AIFF<ul style="list-style-type: none">● Developed by Apple● High Quality (Lossless)● Large File Size | <ul style="list-style-type: none">● MP3 (.mp3)<ul style="list-style-type: none">● Compressed format● Popular for music (MP3 Players)● Well suited for Internet● Ogg Vorbis (.ogg)<ul style="list-style-type: none">● Open Source● Compressed format● Well Suited for Internet● Popular with Gaming Industry |
|---|---|

- **Video resolution** is measured as **width by height** using pixels as units.

Short Name	Name on Device	Video Resolution	File Size (1 minute)
Ultra HD or 4K اعلي دقة	3840 x 2160p	3840 x 2160	320 MB
Full HD	1920 x 1080 / 1080p	1920 x 1080	149MB
HD	1280 x 720 / 720p	1280 x 720	105 MB
WVGA / (Wide SD) اقل دقه	720 x 480 / WVGA	720 x 480	26 MB

Codec

- It **compresses** and **decompresses** (encodes and decodes) the video – with varying degrees of quality depending on the codec. There are two types of codecs: lossy and lossless.
- **Lossy codecs** compress the video **by losing details of the video**. If a video is heavily compressed, the result is poor picture quality.
- **Lossless codecs**, on the other hand, compress the video **without losing details** – so the quality of the original is fully preserved. However, lossless codecs are usually very resource-intensive and the video becomes very large as a result.

1. MP4

MP4 is the **golden mean among the types of video files** when it comes to file size, loading speed, and compression.

Suitable for **streaming**, supports overlay texts, and ensures **lossless compression**. That's why it became widely popular on the **web**. Now, most **websites, social media** platforms, and multimedia players use it as an almost universal standard.

Pros:

- High degree of **compression**,
- Contains **metadata**,
- Versatile, with two extensions (mp4a and mp4v).

Cons:

- **Codec-dependent**,
- MP4 file formats don't protect creators from copyright infringement.

2-MKV

MKV can store captions تعليقات كتابية, audio, and video as a single file making it a popular option for playback of lengthy videos such as movies. Due to the larger file size in comparison to MP however, MKV is more frequently used for downloads and DVD/Blu-Ray playback rather than streaming.

Pros:

- Open-source offers more editing freedom,
- High video quality,
- Supports all codecs.

AVI

5–7 years ago AVI used to be one of the most popular video formats out there. Although it got dethroned over time, the video type still gets credit for its high compatibility. For instance, it's one of the few formats you can run both on Windows, Mac, and Linux devices.

The main issue with AVI is its poor compression as it severely damages the quality of the video. So, AVI files are usually larger than the rest making it a better option for storage, not so much for streaming.

MOV

MOV is known as the “video editing software friendly format” among different file types. It was designed by Apple.

Pros:

- Compatible with most formats.
- Facilitates HD video making.

Cons:

- Large file size.
- Lossy video compression.
- Not compatible with all popular players (programs to play videos).

WEBM

WEBM is a **Google-backed video format** created for web videos. The format supports HTML5 and focuses on keeping clips as small as possible — this way, it won't slow the page down.

Pros:

- **Open-source** and free of royalty licenses.
- **Lossless** compression .
- **Small** file size.

Cons:

- **Less** quality than MP4 .
- Limited support for **mobile devices**.
- **No Mozilla Firefox** support.